

Scalability & Future Plan

Date	11 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited support for scanned PDFs and image-based documents	Users cannot extract text from scanned study materials without OCR.	High
2	Dependence on third-party AI and cloud services (Groq/Firebase)	Service availability may be affected by API limits or network issues.	Medium
3	No dedicated mobile application	Users can only access the platform through a web browser, reducing convenience.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Limited concurrent user support on the current deployment.	Implement auto-scaling, load balancing, and cloud-based deployment to support thousands of concurrent users.
2	Data Storage	Application uses Firebase Firestore with Local JSON storage for development.	Migrate to a fully managed cloud database with automated backups, replication, and scalable storage.
3	Performance	Backend runs on a single server instance and relies on external AI APIs.	Add caching, asynchronous background processing, API optimization, and multiple backend instances for improved

			performance.
4	Security	JWT authentication and Firebase security rules are implemented.	Enhance security with Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), rate limiting, HTTPS enforcement, and regular security audits.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	OCR support for scanned PDFs, Email notifications, Advanced AI analytics	3–6 Months	Improves document processing, user engagement, and learning insights.
Phase 3	Mobile application (Android/iOS), Multi-language support, Real-time collaboration	6–12 Months	Increases accessibility, supports more users, and enables collaborative learning.
Phase 4	AI-powered personalized tutor, Voice-based learning, Cloud auto-scaling and enterprise deployment	12+ Months	Provides an intelligent learning experience, supports large-scale users, and enhances system scalability.